Musings about the Yoneda lemma

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First have a look at the Wikipedia article on the Yoneda lemma. Then have a look at what Mazur has to say about it (referred below). Also, this is what Wikipedia tells us about the Yoneda *embedding*:

> Mapping each object A in $\mathcal{C}$ to its associated hom-functor $h_A = \mathrm{Hom}(A, -)$ and each morphism $f: B \to A$ to the corresponding natural transformation $\mathrm{Hom}(f, -)$ determines a contravariant functor $h_\bullet$ from $\mathcal{C}$ to $\mathbf{Set}^{\mathcal{C}}$, the functor category of all (covariant) functors from $\mathcal{C}$ to $\mathbf{Set}$. One can interpret $h_\bullet$ as a covariant functor:
>
> $h_\bullet : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}^{\mathcal{C}}$.

And now, let me unleash **my understanding** upon the unsuspecting world.

Given any locally small category C, there is a fully faithful contravariant functor (called the Yoneda embedding) from C to the category of (covariant) functors from C to $\mathbf{Set}$.

This contravariant functor can instead be thought of as a covariant functor $h_\bullet : C^{\mathrm{op}} \to [C, \mathbf{Set}]$, where $[A, B]$ is the category of (covariant) functors from A to B.

$h_\bullet$ consists of an object map $h_{\bullet_{\mathrm{obj}}} : \mathrm{obj}(C^{\mathrm{op}}) \to \mathrm{obj}([C, \mathbf{Set}])$, defined as

$$\begin{aligned} A &\mapsto h_A \\ &= \mathrm{Hom}_C(A, -) \\ &= (\forall B \in \mathrm{obj}(C), B \mapsto \mathrm{Hom}_C(A, B)), \end{aligned}$$

and a morphism map $h_{\bullet_{\mathrm{mor}}} : \mathrm{Hom}_{C^{\mathrm{op}}}(A, B) = \mathrm{Hom}_C(B, A) \to \mathrm{Nat}(h_A, h_B)$, defined as

$$f \mapsto (\forall X \in \mathrm{obj}(C), X \mapsto (\forall g \in \mathrm{Hom}(A, X), g \mapsto g \circ f))$$

If the morphism map $h_{\bullet_{\mathrm{mor}}}$ seems like a lot to unpack, check the type at each stage. Given $f \in \mathrm{Hom}_C(B, A)$, $h_{\bullet_{\mathrm{mor}}}$ needs to produce a natural transformation from h_A to h_B. The natural transformation should convert $X \in \mathrm{obj}(C)$ to a morphism (in $\mathbf{Set}$) from $\mathrm{Hom}(A, X)$ to $\mathrm{Hom}(B, X)$. Given $g \in \mathrm{Hom}(A, X)$, the produced element of $\mathrm{Hom}(B, X)$ is $g \circ f$.

Now, let us discuss the **fullness and faithfulness** of $h_\bullet$. The Yoneda lemma tells us that given a functor F from C to $\mathbf{Set}$,

$$\forall A \in \mathrm{obj}(C), \mathrm{Nat}(h_A, F) \cong F(A).$$

Now, in the case when F is also, some hom-functor h_B, this reduces to

$$\forall A, B \in \mathrm{obj}(C), \mathrm{Nat}(h_A, h_B) \cong h_B(A) = \mathrm{Hom}_C(B, A) = \mathrm{Hom}_{C^{\mathrm{op}}}(A, B),$$

which is *exactly* what we needed! A one-to-one correspondence between hom-sets in C^{op} (the domain of $h_\bullet$), and $\mathrm{Nat}(h_A, h_B)$, i.e., the corresponding hom-sets in $[C, \mathbf{Set}]$ (the codomain of $h_\bullet$).

$h_\bullet$ being a fully faithful functor means that C^{op} is isomorphic to a full subcategory of $[C, \mathbf{Set}]$. The objects of this subcategory are exactly the hom-functors on C. Such functors are called representable functors.

References:

- https://en.wikipedia.org/wiki/Yoneda_lemma
- *When is one thing equal to some other thing?*, Barry Mazur (Section 14, pg. 18-19)
- https://mathoverflow.net/q/3184 (Very nice answers and resources here).
- *Categories for the Working Mathematician*, Saunders MacLane (Chapter I, Section 3, "Functors") [Contains discussion about fullness and faithfulness of functors]
- https://en.wikipedia.org/wiki/Hom_functor
- https://en.wikipedia.org/wiki/Representable_functor